

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

NAME: M A SURYA KIRAN

REG.NO: 192411236

COURSE NAME: CSA1603

COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

Assessment Tool Weightage: 35%

QUESTIONS: 05

Total Marks:50

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding (2.5)	Constraint Analysis (2.5)	Solution Development (2.5)	Application of Concepts (1.5)	Justification (1)	Total(10)
<i>Weightage</i>	25%	25%	25%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I M A SURYA KIRAN (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

M A SURYA KIRAN

Signature of the student:

RUBRICS FOR CALCULATION:

**Constraint-Based Problem Solving
Assessment Rubric**

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding ; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving (6 Marks) (BL4–BL5) – Reframed with New Scenarios

Q1. Dataset: Hospital Patient Records

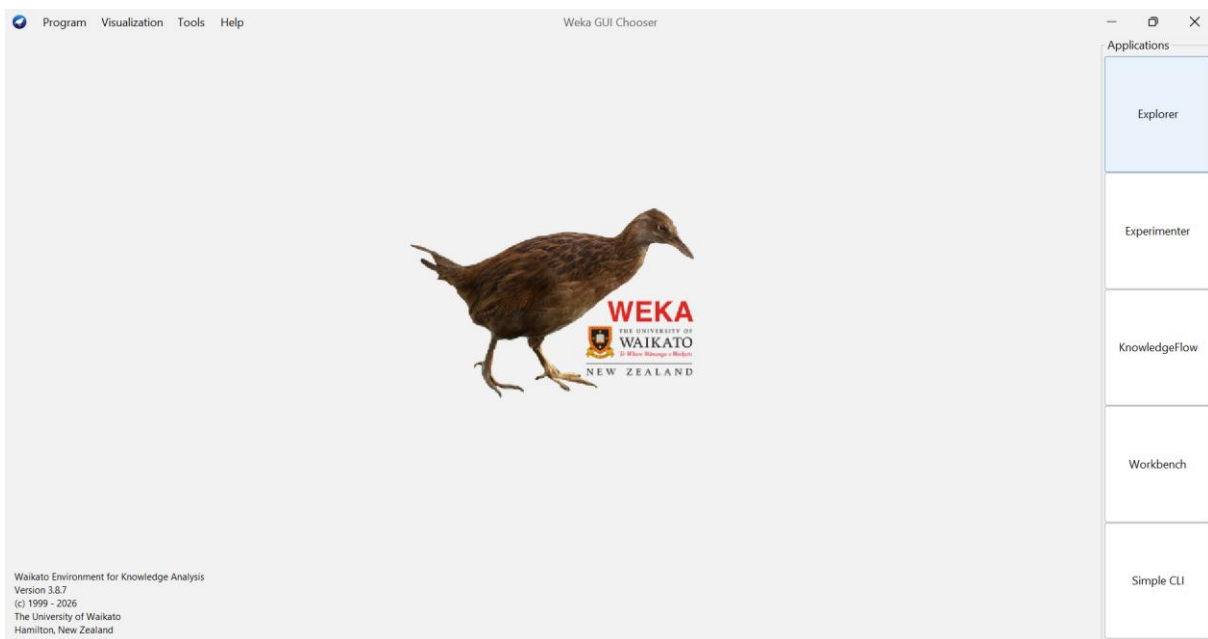
PID Symptoms Observed

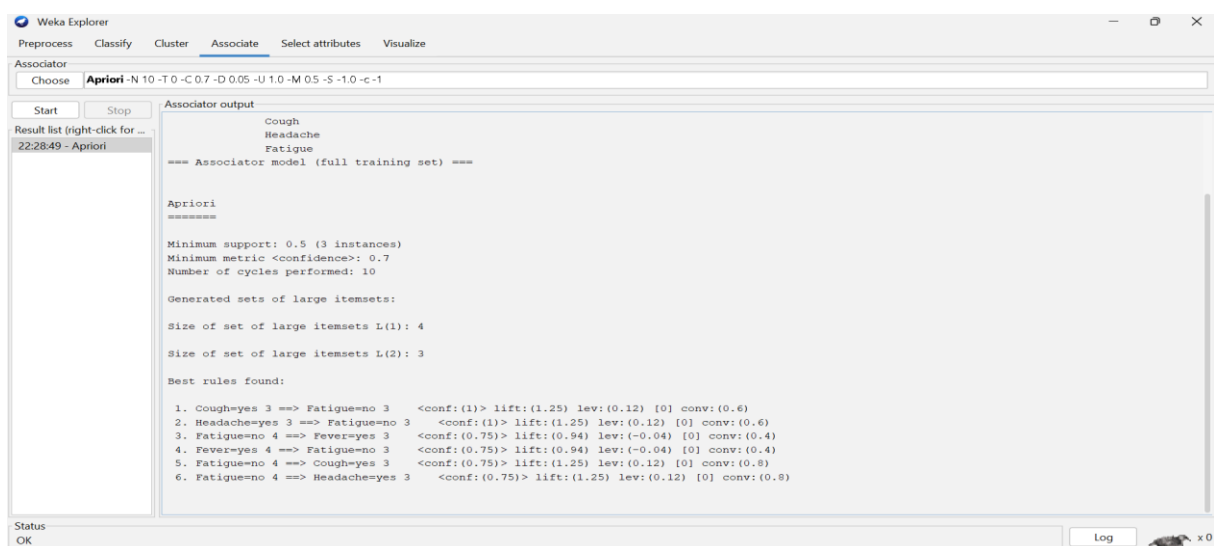
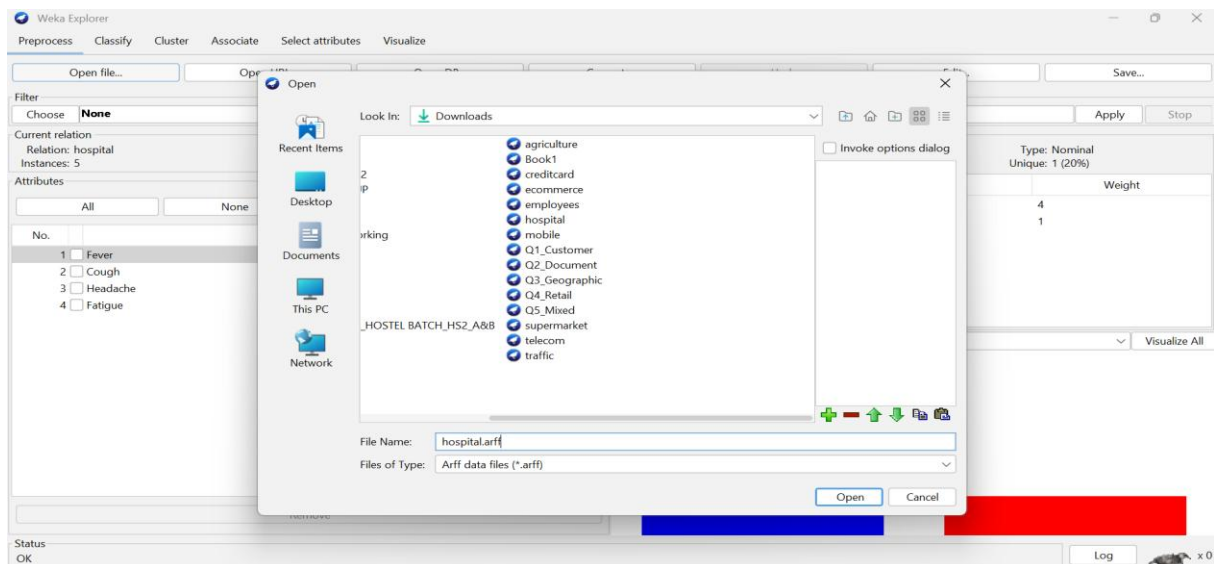
- P1 Fever, Cough
- P2 Fever, Headache
- P3 Cough, Headache
- P4 Fever, Cough, Headache
- P5 Fever, Fatigue

Question:

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include **"Fever"**. Identify the valid association rules and justify why they satisfy the given constraints. (BL4 – 1.2 Marks)

ANSWER:





Conclusion

The Fever-only itemset is frequent, but there is no meaningful Fever-containing association rule satisfying both support $\geq 50\%$ and confidence $\geq 70\%$ from this dataset.

This is an important observation for your answer: WEKA may produce no valid Fever-based rule under these exact constraints.

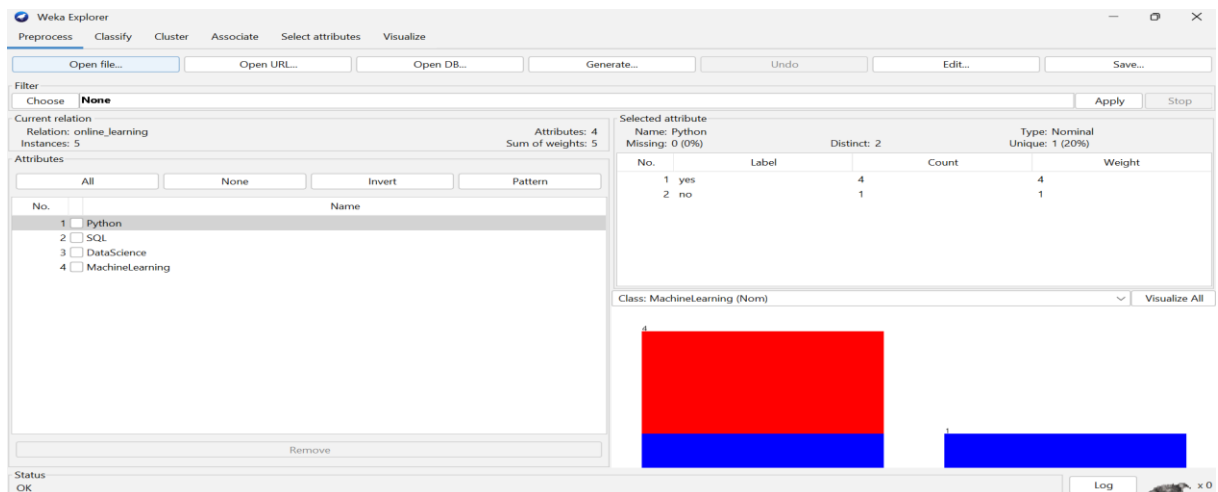
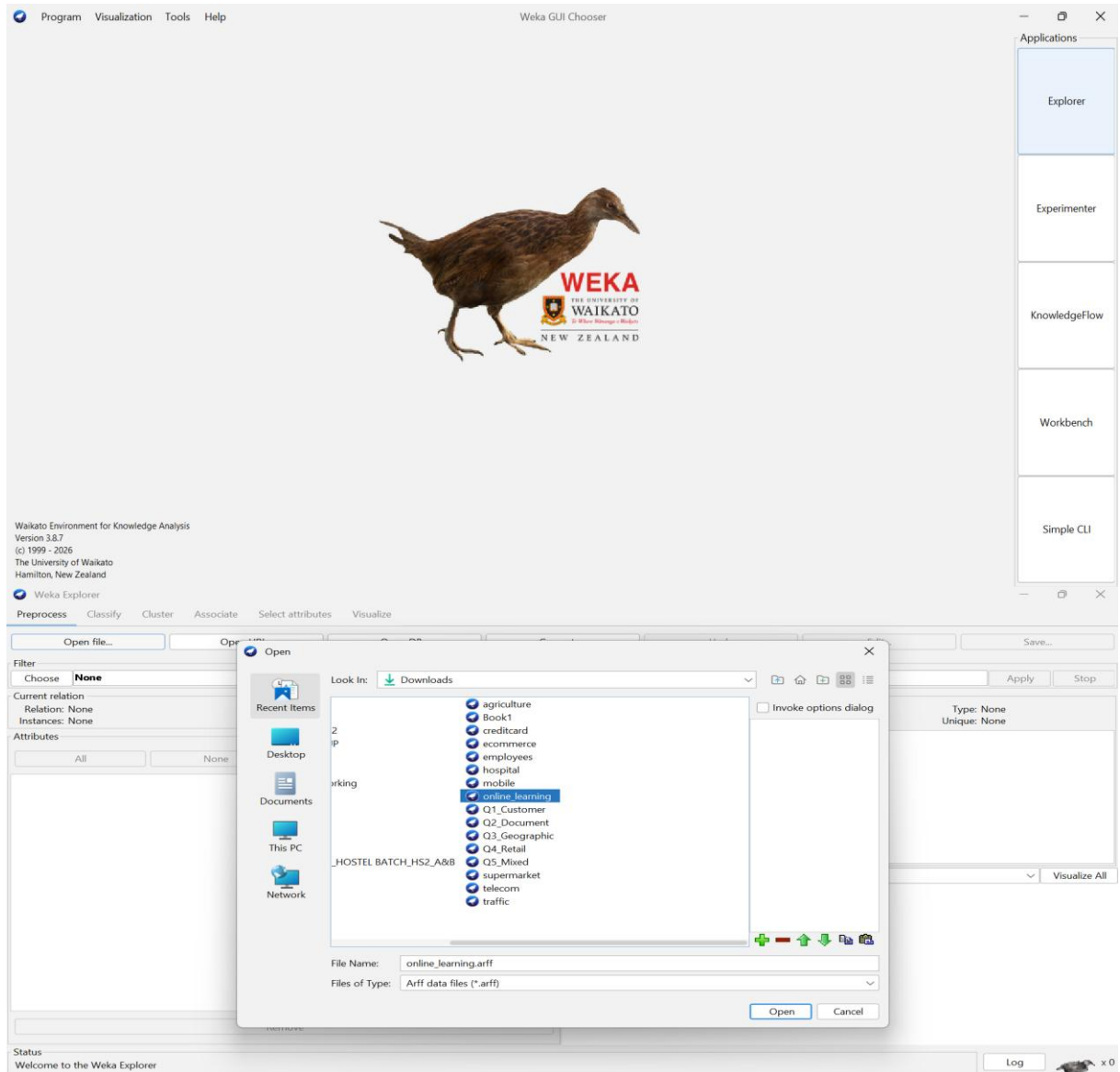
Q2. Dataset: Online Learning Platform

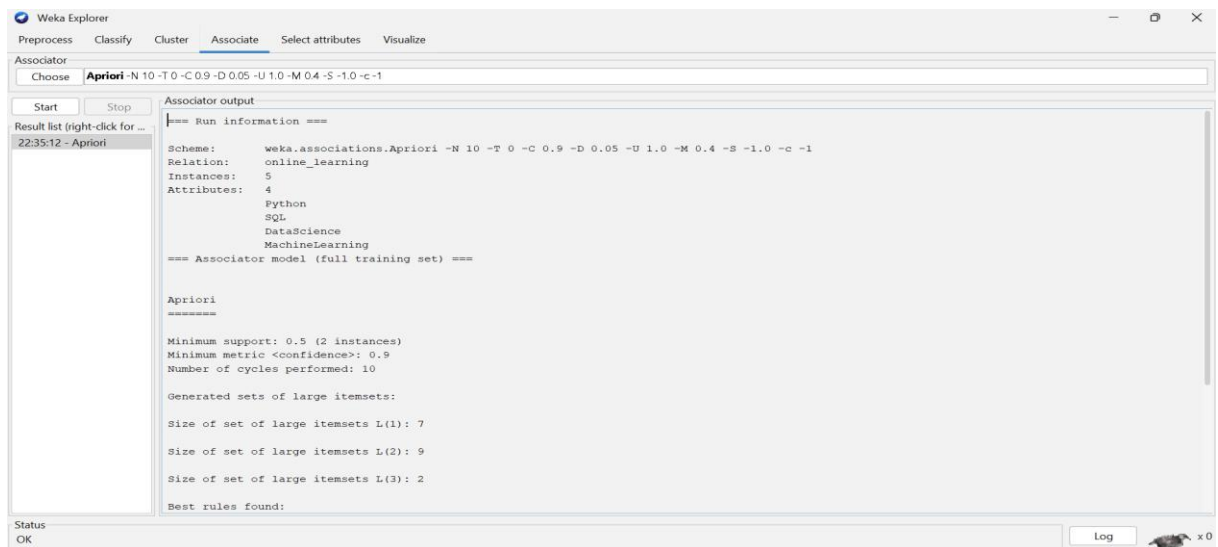
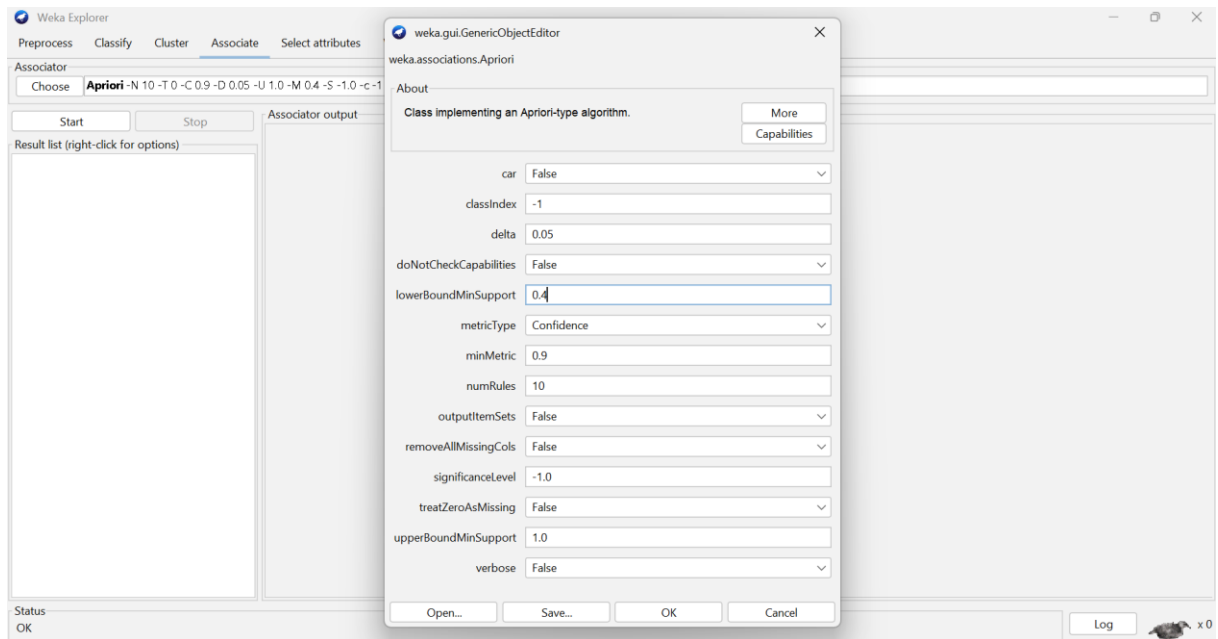
Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining. (*BL4 – 1.2 Marks*)

ANSWER:







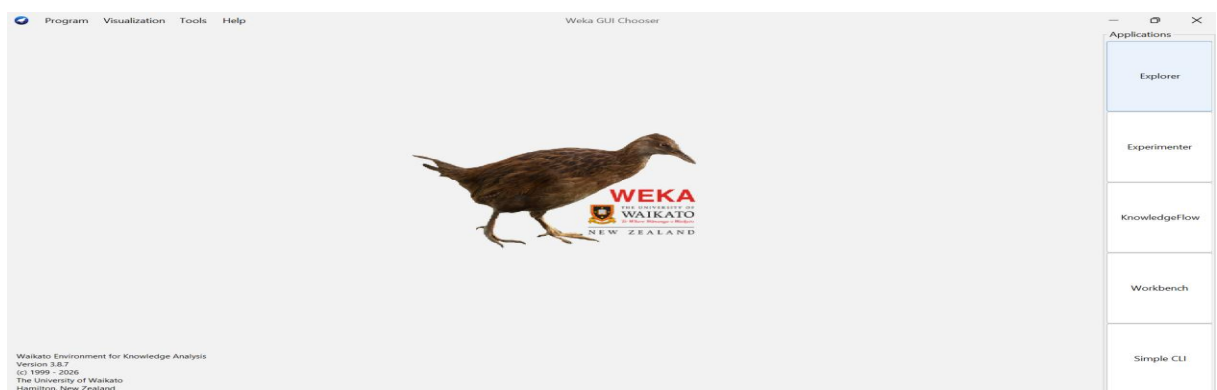
Q3. Dataset: Library Borrowing Records

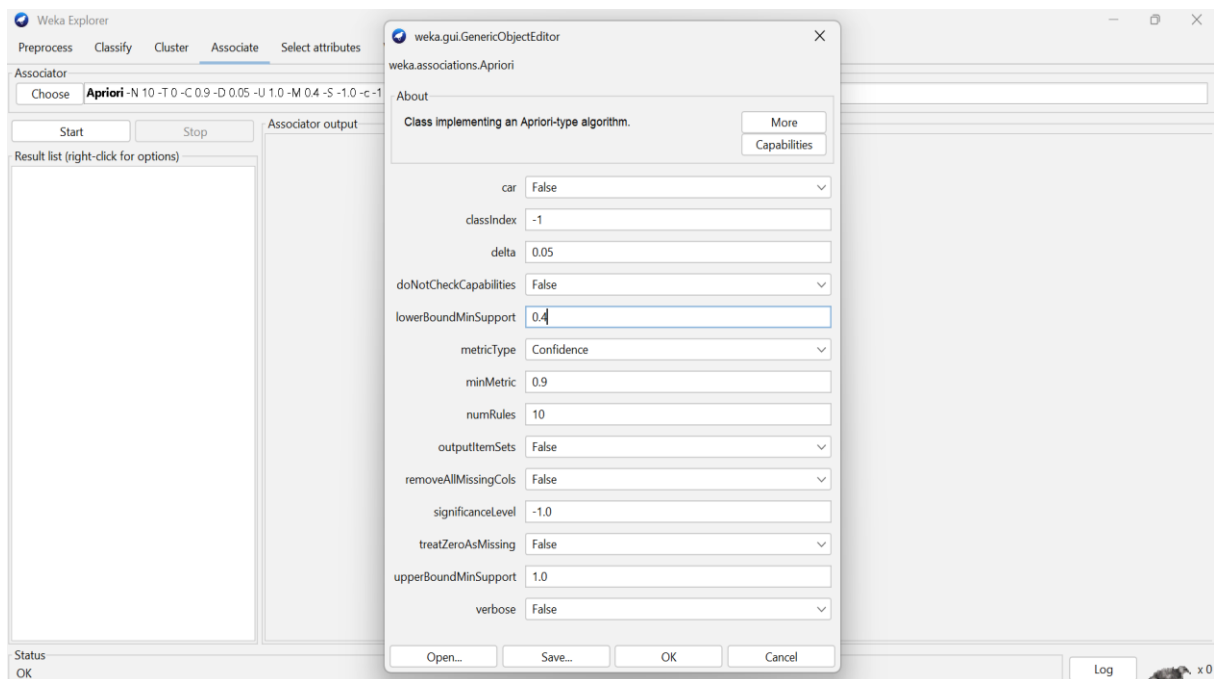
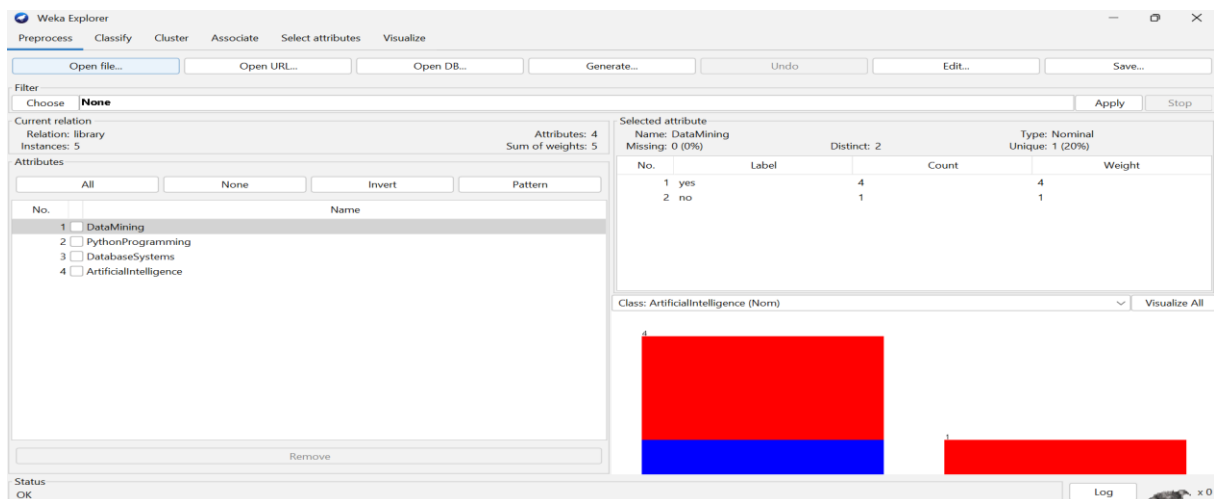
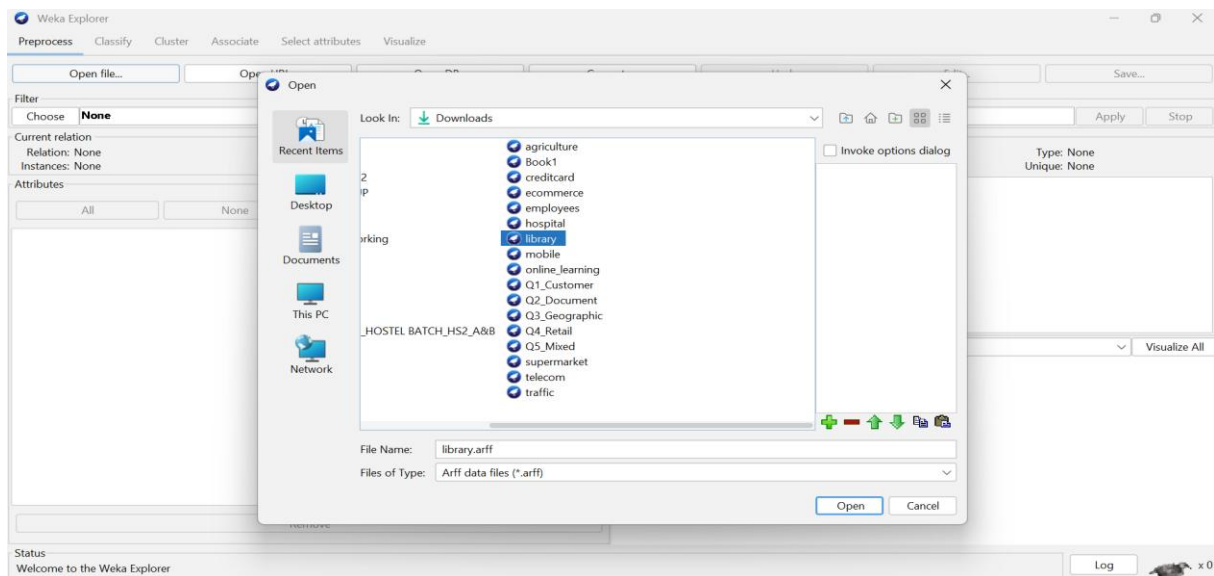
Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

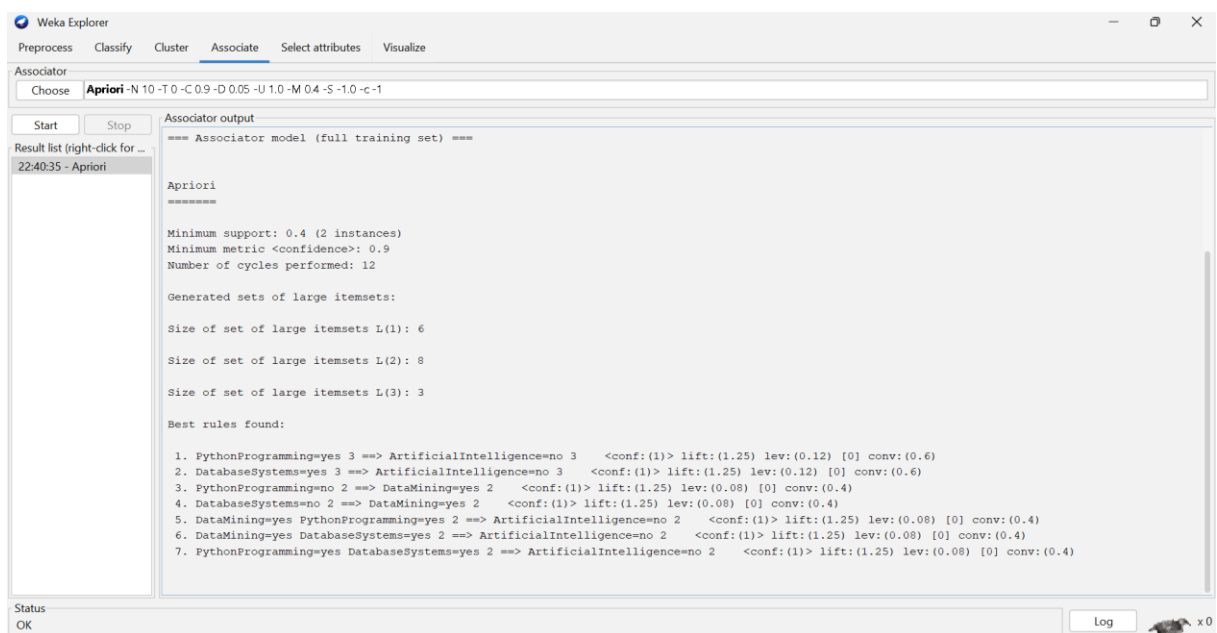
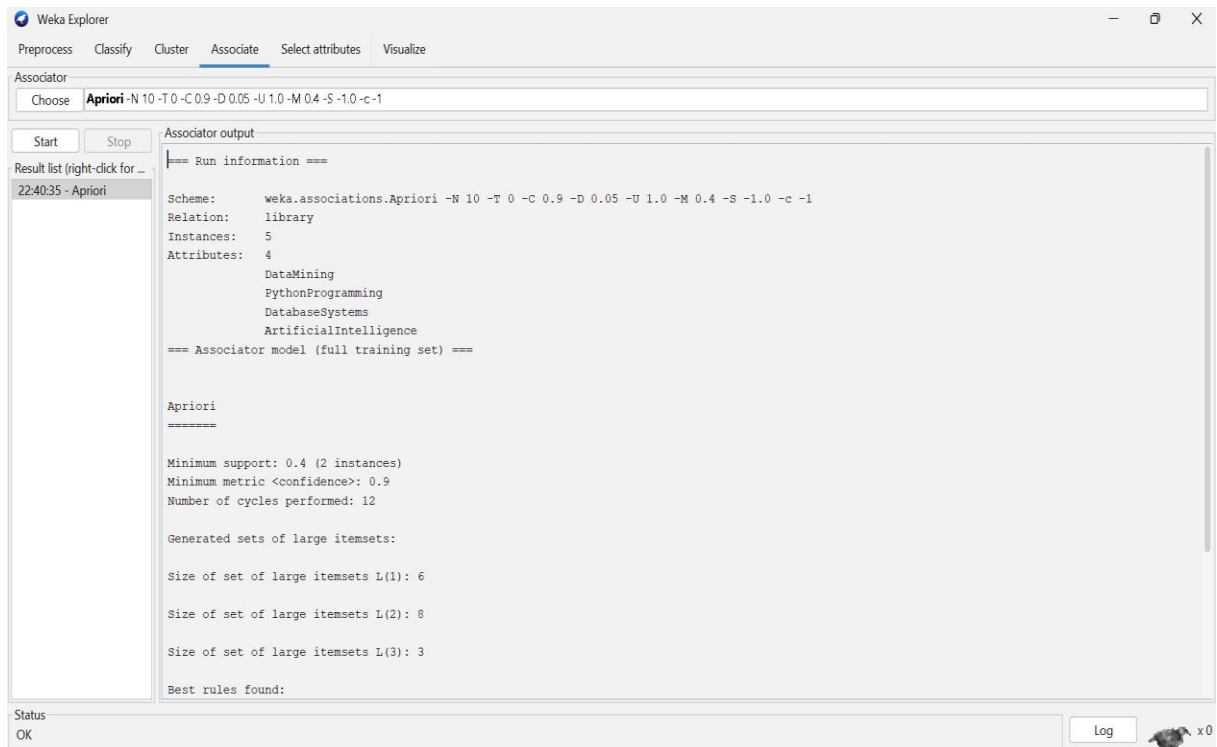
Question:

Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation. (*BL5 – 1.2 Marks*)

ANSWER:







Q4. Dataset: University Course Hierarchy

Student ID

Courses Taken

S1

Engineering, Computer Science, Machine Learning

Student ID

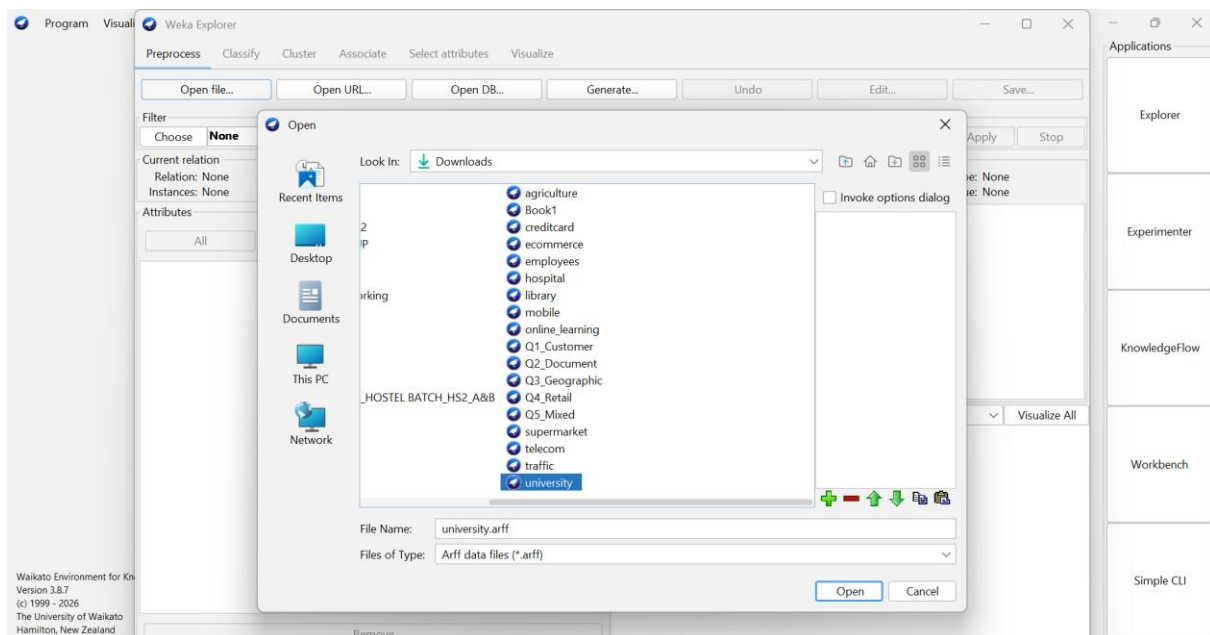
Courses Taken

S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the "**Computer Science**" level. Explain how the course hierarchy affects the discovery of association rules. (BL5 – 1.2 Marks)

ANSWER:



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation:
Relation: university
Instances: 5

Attributes: 4 Sum of weights: 5

Attributes: All None Invert Pattern

No.	Name
1	☒ Engineering
2	☐ ComputerScience
3	☐ MachineLearning
4	☐ InformationTechnology

Remove

Status: OK

Selected attribute:
Name: Engineering
Missing: 0 (0%) Distinct: 1 Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	yes	5	5
2	no	0	0

Class: InformationTechnology (Nom) Visualize All

Log x 0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes

Associator: Choose **Apriori** -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1

Start Stop Associator output

Result list (right-click for options)

Status: OK

weka.gui.GenericObjectEditor

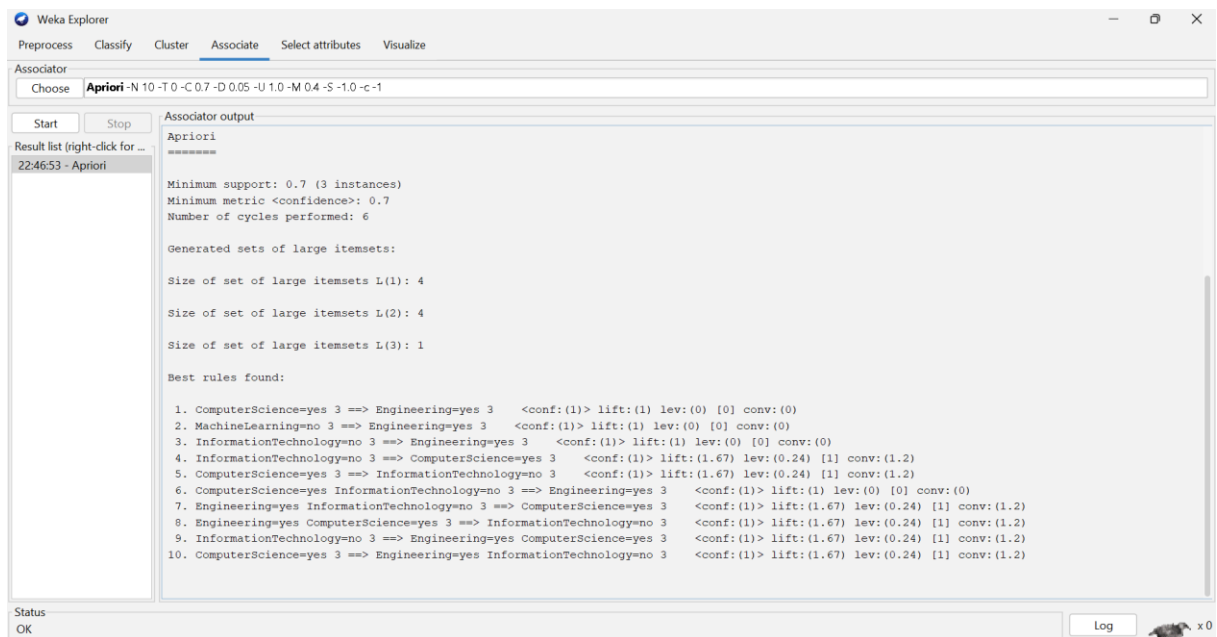
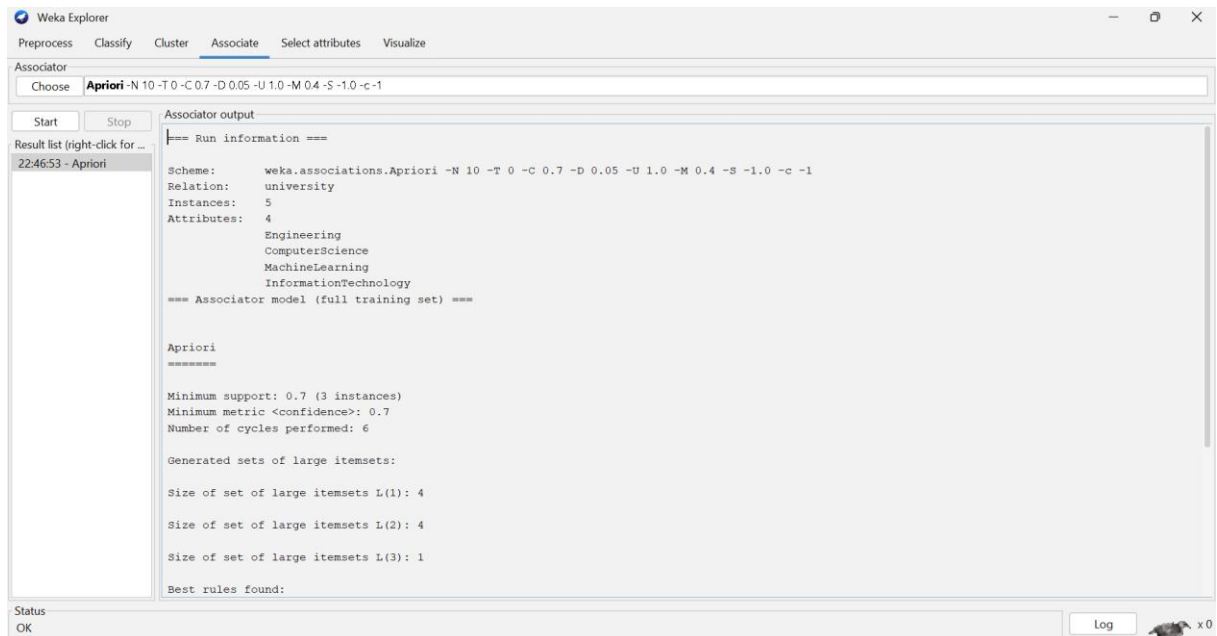
weka.associations.Apriori

About: Class implementing an Apriori-type algorithm. More Capabilities

car	False
classIndex	-1
delta	0.05
doNotCheckCapabilities	False
lowerBoundMinSupport	0.4
metricType	Confidence
minMetric	0.7
numRules	10
outputItemSets	False
removeAllMissingCols	False
significanceLevel	-1.0
treatZeroAsMissing	False
upperBoundMinSupport	1.0
verbose	False

Open... Save... OK Cancel

Log x 0



Q5. Dataset: Insurance Customer Records

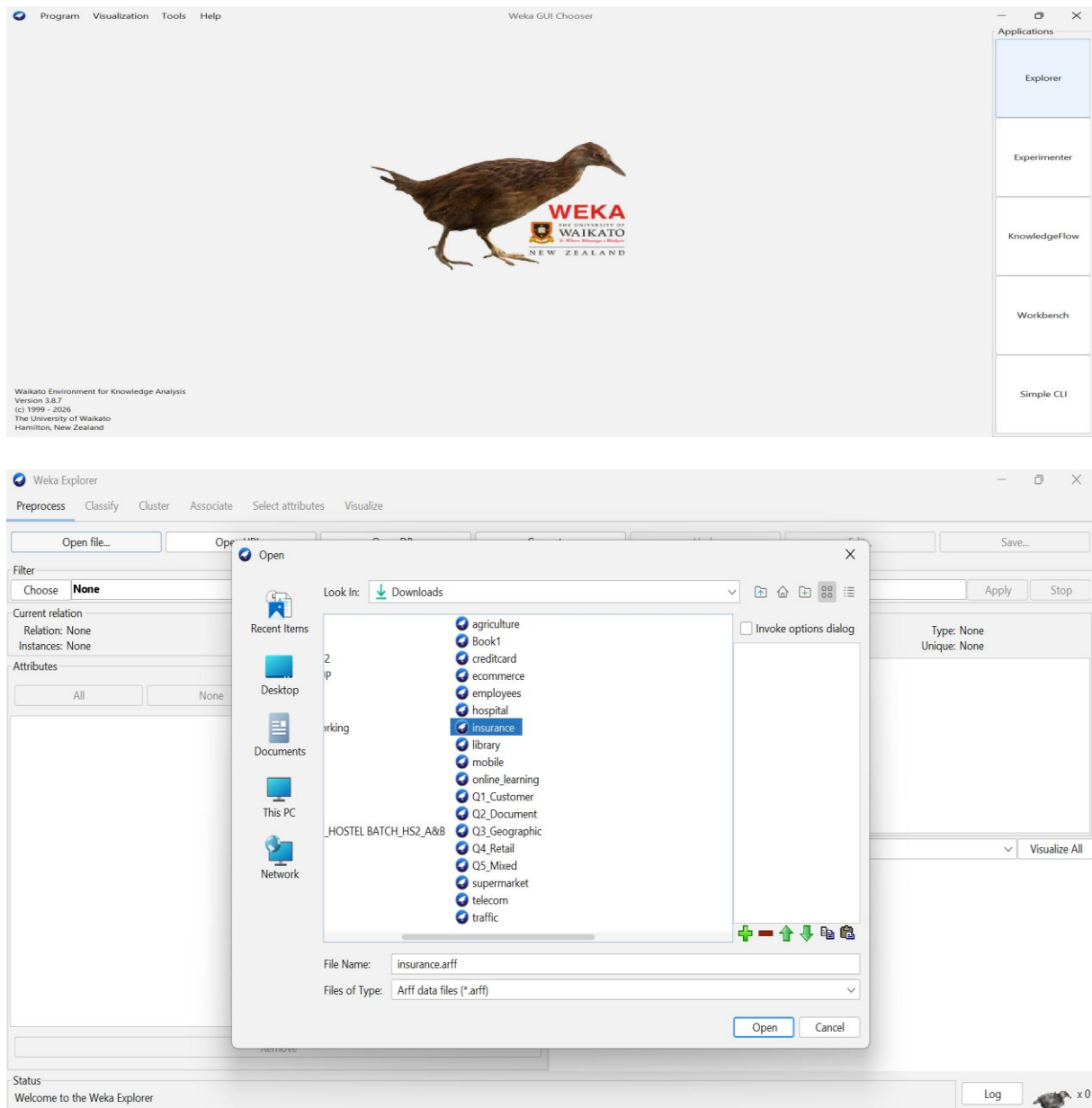
CID Age Group Income Level Policy Purchased

C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:

Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies. (BL4 – 1.2 Marks)

ANSWER:



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation
Relation: insurance
Instances: 5 Attributes: 3 Sum of weights: 5

Attributes

All None Invert Pattern

No.	Name
1	☒ AgeGroup
2	☐ IncomeLevel
3	☐ Policy

Remove

Status: OK

Selected attribute
Name: AgeGroup
Missing: 0 (0%) Distinct: 3 Type: Nominal
Unique: 2 (40%)

No.	Label	Count	Weight
1	Young	3	3
2	Middle	1	1
3	Senior	1	1

Class: Policy (Norm) Visualize All

Weka Explorer

Preprocess Classify Cluster Associate Select attributes

Associator: Choose **Apriori** -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1

Start Stop Associator output

Result list (right-click for options)

Status: OK

weka.gui.GenericObjectEditor

weka.associations.Apriori

About

Class implementing an Apriori-type algorithm. More Capabilities

car	False
classIndex	-1
delta	0.05
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lowerBoundMinSupport	0.4
metricType	Confidence
minMetric	0.7
numRules	10
outputItemSets	False
removeAllMissingCols	False
significanceLevel	-1.0
treatZeroAsMissing	False
upperBoundMinSupport	1.0
verbose	False

Open... Save... OK Cancel

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

22:53:42 - Apriori

Associator output

```
=== Run information ===

Scheme:      weka.associations.Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1
Relation:    insurance
Instances:   5
Attributes:  3
              AgeGroup
              IncomeLevel
              Policy

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.4 (2 instances)
Minimum metric <confidence>: 0.7
Number of cycles performed: 12

Generated sets of large itemsets:

Size of set of large itemsets L(1): 5

Size of set of large itemsets L(2): 2

Best rules found:

1. Policy=Health 2 ==> IncomeLevel=High 2   <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
2. IncomeLevel=High 2 ==> Policy=Health 2   <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
```

Status
OK

Log x 0

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.4 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

22:53:42 - Apriori

Associator output

```
Instances:   5
Attributes:  3
              AgeGroup
              IncomeLevel
              Policy

=== Associator model (full training set) ===

Apriori
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1. Policy=Health 2 ==> IncomeLevel=High 2   <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
2. IncomeLevel=High 2 ==> Policy=Health 2   <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
3. Policy=Vehicle 2 ==> IncomeLevel=Medium 2 <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
4. IncomeLevel=Medium 2 ==> Policy=Vehicle 2 <conf:(1)> lift:(2.5) lev:(0.24) [1] conv:(1.2)
```

Status
OK

Log x 0